Temperature and precipitation from ice extents during the Last Glacial Maximum: a scalable, transient, surrogate-based inversion

F.

Abstract. Past climate can be constrained by mapped glacier extents, but the link from ice margin to temperature and precipitation is neither direct nor unique. Here we infer spatially varying changes in temperature and precipitation from the observed ice extent of the Last Glacial Maximum (LGM) in two steps. First, we invert the mapped extent for a spatially variable equilibrium line altitude (ELA), the elevation where a glacier gains as much ice as it loses, using a regularized fixed-point iteration under a transient climate forcing. This first steps starts with the shallow ice approximation, which is computationally cheap and gives a first set of ELA fields. These fields guide forward runs of the Instructed Glacier Model, (IGM) which solves the higher-order Blatter–Pattyn equations. We then use these IGM runs to train a neural-network surrogate. The surrogate reduces an inversion from hours to seconds, which makes it possible to repeat the inversion for many mass-balance gradients and thereby obtain a range of ELA fields that can explain the observed ice extents. Second, we convert the ELA into temperature and precipitation changes with a positive-degree-day model and a Clausius–Clapeyron relation, and we propagate the uncertainty of the lapse rate, the seasonality and the modern ELA. We illustrate how the method works by applying it to the South Island of New Zealand, where the ELA varies strongly across the mountains. We show how such result can be used to quantify a cooling of 4.8 to $6.5\,\mathrm{K}$ at three key sites, with 29 to $37\,\%$ less precipitation if precipitation follows the Clausius–Clapeyron relation, and a standard deviation of about $0.8\,\mathrm{K}$. The method can be reused for other regions with mapped ice extents.

1 Introduction

The Last Glacial Maximum (LGM), about $26{,}500$ to $19{,}000$ years ago, is the last period when ice sheets and glaciers reached their greatest combined extent, leading to a lowering global sea level by about $125\,\mathrm{m}$ (Clark et al., 2009). Quantifying the cooling that accompanied this change has been a long-standing goal. Recent global reconstructions that combine marine geochemical proxies with climate models suggest that Earth was on average about $6.1^{\circ}\mathrm{C}$ colder than in pre-industrial times, with extreme cooling in the polar regions and in some continental interiors (Tierney et al., 2020). On land, temperature and moisture patterns have also been quantified from subfossil pollen assemblages. Pollen records indicate widespread cooling and drying, driven by greenhouse gas and orbital changes and modified locally by atmospheric circulation (Bartlein et al., 2011; Prentice et al., 2017). Coupled atmosphere–ocean models complement these reconstructions. The latest generation reproduces large-scale features such as polar amplification, poleward shifts of the jet streams and changes in the Atlantic circulation, and it suggests a wider range of global cooling and overall drier conditions (Kageyama et al., 2021).

It is thought, however, that models tend to underestimate regional winter cooling and local precipitation changes (Cleator et al., 2020; Kageyama et al., 2021). Concerns have also been raised about biases associated with the difficulty of accounting for

changes in CO_2 when pollen is used as a climate proxy (Prentice and Harrison, 2009; Prentice et al., 2017, 2022). Furthermore, pollen records are inherently limited in fully glaciated regions, where temperature and precipitation changes may have been greatest, because the absence of vegetation precluded pollen deposition. Similarly, terrestrial inferences from the marine proxy records relies on model-based extrapolation for marine to terrestrial regions. As a consequence, substantial uncertainties remain in mountain ranges, where temperature and precipitation gradients are steepest.

Glacier extent provides an alternative archive of past climate that can help address some of these limitations. Glaciers advanced in response to climatic changes, leaving moraines that record where and, in some cases, when these advances occurred. Mapped glacier extents are now available for many regions (Ehlers et al., 2011). Although glaciers did not all reach their maximum extent simultaneously, most culminated during the LGM (Clark et al., 2009; Hughes et al., 2013; Ehlers et al., 2011). For example, high-resolution reconstructions in New Zealand reveal several distinct culminations of valley glaciers (Barrell et al., 2011), while chronological and morphostratigraphic compilations from the European Alps provide detailed constraints on glacier advance and retreat during the LGM (Wirsig et al., 2016; Kamleitner et al., 2026). These observations imply that mapped maximum extents can be compared with the maximum extent simulated over a period, rather than requiring all glaciers to have reached their maximum at a single time. Such data can, in principle, constrain the climate conditions that produced the ice, but doing so requires an ice-flow model that links climate to glacier extent.

A first way to use ice-flow models is to run them forward. In this configuration, climate is prescribed, either as temperature and precipitation offsets applied to a present-day climate (Golledge et al., 2012; Seguinot et al., 2018) or from downscaled climate simulations (Jouvet et al., 2023), and the simulated ice is then compared with the mapped extent. This approach has been invaluable, because it shows how ice fields grew and shrank in response to changes in temperature and precipitation. It has, however, two limitations. Climate models have a coarse resolution, which introduces downscaling uncertainty in mountain regions. Furthermore, the results depend on parameters and empirical assumptions of the ice and climate models that are hard to constrain.

Inverse modelling offers a second alternative. Instead of prescribing the climate that prevailed, it uses the geological footprint directly and adjusts the mass balance until the model reproduces the mapped extent. Višnjević et al. (2018, 2020) and Lleshi et al. (2025) introduced inversion schemes that recover a field of equilibrium line altitude (ELA), the elevation where annual accumulation equals ablation, which can then be converted into temperature and precipitation. Both steps are non-unique. Different combinations of ELA and mass-balance gradients produce the same extent, and one ELA is consistent with many pairs of temperature and precipitation. This is the equifinality of the problem, and the result should therefore be a range and not a single value. Producing such a range is possible but requires many inversions, which has been a challenge because of its computationally expensive cost.

Here we build on the recent efforts of inverting a mapped ice extent into an ELA field. We use a regularized fixed-point iteration under transient climate forcing (Višnjević et al., 2018), without assuming that the ice is in steady state (Lleshi et al., 2025). We begin with the shallow-ice approximation (SIA), which is computationally efficient and provides initial ELA fields. These fields guide the selection of a limited number of forward simulations with the Instructed Glacier Model (Jouvet and Cordonnier, 2023), which solves the higher-order Blatter–Pattyn equations. We use the results of these simulations to train

a neural-network surrogate, which is then incorporated into the same fixed-point iteration inversion scheme. By guiding the selection of training simulations with the SIA-derived ELA field, we limit the amount of training data required while enabling hundreds of inversions across a range of mass-balance gradients that cannot be uniquely constrained by the mapped extent alone. This ensemble of ELA fields is then reduced to the inversions that reproduce the mapped extent to within a tolerance. After each inversion stage, we assess whether additional training data are needed by running the higher-order models, then generate new simulations where necessary, and retrain the surrogate. We repeat this procedure until the surrogate agrees with the ice-flow model where the inversions end. Second, we translate the inferred ELA field into temperature and precipitation changes using a positive degree-day (PDD) model and the Clausius–Clapeyron relationship, propagating uncertainties from both the inversion and the climate reconstruction. We illustrate the method using the mapped Last Glacial Maximum (LGM) ice extent on the South Island of New Zealand. The method is designed to be applicable to other regions where LGM ice extents have been mapped.

2 Methods

In this section, we describe the forward models and inversion method.

2.1 Mass conservation and surface mass balance

Ice thickness $h(x, y, t)$ over a bed $b(x, y)$, with surface $s = b + h$, obeys

$$\frac{\partial h}{\partial t} = m - \nabla \cdot \mathbf{Q}, \tag{1}$$

where $\mathbf{Q}$ is the depth-integrated flux and m the surface mass balance. We parameterise m by a two-gradient law about the ELA,

$$m(s) = \begin{cases} \min\left[\beta_{\mathrm{acc}}(s - \mathrm{ELA}), c\right], & s > \mathrm{ELA}, \\ \beta_{\mathrm{abl}}(s - \mathrm{ELA}), & s \leq \mathrm{ELA}, \end{cases} \tag{2}$$

with a cap c on the accumulation rate. Separate gradients are needed because a single one conflates the temperature (melt) and precipitation signals.

2.2 Ice-flow models: shallow-ice and higher-order, steady and transient

Ice thickness evolves by the mass conservation equation (1), which we solve with two ice-flow models of different complexity. The shallow-ice approximation (SIA) (Hutter, 1983) is used for the baseline inversion and to provide starting ice states for generating training data. The Instructed Glacier Model (IGM) solves a higher-order approximation that is integrated in time. We use the latter solution to train a surrogate using neural networks (**?**), so the inversion is feasible (**?**). Both models share the same flow-law coefficients and the same treatment of the lithosphere.

2.2.1 Shallow-ice approximation

Under the shallow-ice approximation (Hutter, 1983), valid where the ice is much wider than it is thick, the flux is purely local and diffusive, and mass conservation reduces to a non-linear diffusion equation,

$$\mathbf{Q} \;=\; -D\,\nabla s, \qquad D \;=\; \Big[\underbrace{\frac{2A}{n+2}(\rho g)^n h^{n+2}}_{\text{deformation}} + \underbrace{C_s(\rho g)^n h^n}_{\text{sliding}}\Big]\,|\nabla s|^{n-1}, \tag{3}$$

with Glen exponent $n = 3$, rate factor A, ice density ρ, gravity g and sliding coefficient C_s. We integrate Eq. (1) with the SIA and a spatially variable mass balance, through a spatially variable ELA and fixed mass balance gradients. We do not vary the mass balance in time.

2.2.2 Higher-order model

The SIA drops the longitudinal and lateral stresses. These can matter for the thickness and the velocity of mountain ice caps, and we therefore use IGM, which solves the Blatter–Pattyn equations,

$$\frac{\partial}{\partial z}\Big(\eta\frac{\partial u}{\partial z}\Big) + \frac{\partial}{\partial x}\Big[2\eta\Big(2\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\Big)\Big] + \frac{\partial}{\partial y}\Big[\eta\Big(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\Big)\Big] = \rho g\,\frac{\partial s}{\partial x}, \qquad \eta = \tfrac{1}{2}A^{-1/n}\dot{\varepsilon}_e^{(1-n)/n}, \tag{4}$$

and the same for v, as an energy minimisation in three dimensions (**?**). The basal sliding follows a Weertman law $\tau_b = \tau_{\mathrm{ref}}(u_b/u_{\mathrm{ref}})^{1/p}$ with exponent $p = 3$.

IGM also integrates Eq. (1) in time. Here the mass balance is prescribed through a spatially and temporally variable ELA, such that

$$\mathrm{ELA}(x,y,t) \;=\; E(x,y) + \Delta E\,\big[1 - s(t)\big], \tag{5}$$

where $E(x,y)$ is the LGM ELA and $s(t)$ is an index, often called glacial index, that scales a climate record (Greve, 1997; Niu et al., 2019; Scherrenberg et al., 2023; **?**; Leger et al., 2025; **?**), and ΔE is a prescribed amplitude.

2.2.3 Comparison of the two models

The two models use the same flow-law and sliding coefficients. These are $A = 78\ \mathrm{MPa}^{-3}\,\mathrm{yr}^{-1}$, $p = 3$ and $\tau_{\mathrm{ref}} = 0.0464$ MPa at $u_{\mathrm{ref}} = 1\ \mathrm{m\,yr}^{-1}$, which in the SIA corresponds to $C_s = u_{\mathrm{ref}}/\tau_{\mathrm{ref}}^3 \approx 1.0\times10^{-14}\ \mathrm{Pa}^{-3}\,\mathrm{m\,yr}^{-1}$. The functional form of the sliding is not identical. For a basal shear stress $\tau_b = \rho g h|\nabla s|$ the Weertman law of IGM gives a sliding flux that grows as h^{p+1}, whereas the two-term SIA form grows as h^n. We tested this and found that the effect is negligible.

Note that for the same ELA field and the same transient forcing, the SIA and IGM solutions give a very similar ice extent when using a 2 km grid spacing. The two agree within 5% in each of six bands across the strike of the mountains. The longitudinal stresses that the SIA neglects therefore do not control the extent we fit. The comparison also shows that the

difference between the SIA and IGM inversions that we report below comes mainly from the transient state of the ice in the inversion and not from the flow model. We decided to use IGM it computes higher order ice physics.

2.2.4 Flexure of the lithosphere

For both models the bed is not rigid. Under an ice load the lithosphere deflects as a thin elastic plate,

$$D_f \nabla^4 w + \rho_m g\, w \;=\; \rho_i g\, h, \qquad D_f = \frac{E T_e^3}{12(1-\nu^2)}, \tag{6}$$

which is algebraic in the wavenumber domain, $\hat{w} = (\rho_i/\rho_m)\hat{h}\big/\big[D_f k^4/(\rho_m g)+1\big]$, and can be solved spectrally. At $k = 0$ this recovers Airy equilibrium, $w = (\rho_i/\rho_m)h$, while shorter-wavelength loads are carried by plate strength. The transition is set by the flexural parameter $\alpha_f = (D_f/\rho_m g)^{1/4}$. Here we use an elastic thickness $T_e = 15$ km, Young's modulus $E = 100$ GPa, Poisson's ratio $\nu = 0.25$ and mantle density $\rho_m = 3300\ \mathrm{kg\,m^{-3}}$, and the deflection is recomputed as the ice evolves.

2.3 A neural-network surrogate of the transient model

A transient run of the higher-order IGM (Blatter–Pattyn) over a time window takes minutes to an hour, depending on the grid and the GPU. One inversion with IGM needs tens of such runs, and an ensemble of inversions over the mass-balance gradients, the forcing amplitude and the random starts needs hundreds of inversions. We therefore train a neural network to emulate the transient higher-order model, as in Lleshi et al. (2025), so that it can replace IGM as the forward model of the inversion. The network is a U-Net, an encoder and decoder with skip connections. Its inputs are six maps, namely the ELA field E, the bed, the domain mask and three maps that hold the constants β_{abl}, β_{acc} and ΔE. Its outputs are the maximum thickness reached over the run and a logit for the presence of ice, from which the mask M_{mod} is obtained. One network is trained for a given region, with a fixed grid, forcing index $s(t)$, time window and starting ice. The surrogate is never used on its own: the inversions it produces are checked against the higher-order IGM, as described with the inversion (Sect. 2.4.4).

The training data are complete runs of the higher-order IGM over the time window, forced by Eq. (5) and started from the ice of the SIA. The target is the maximum thickness over the part of the run used in the comparison with the mapped extent. The runs are generated in rounds. In the first round, a reference ELA field, for example one obtained with a SIA inversion, is perturbed by a shift, a change of scale, a tilt and a correlated random field, and the gradients and the amplitude are drawn over their plausible ranges. The inversion generally ends in a region of this space that the first round samples poorly. In the second round we therefore perturb fields that already fit the mapped extent, and in the third round we may add localised ELA patches, placed where the surrogate and IGM disagree and along the margin of the mapped ice. Adding runs where the optimiser ends up is a form of active learning. A guard aborts runs that flood the domain, whose extent exceeds a multiple of the mapped one, or that put more than a threshold number of cells of ice over the sea, and the aborted runs are not used.

We use a smooth-L_1 loss on the thickness plus 0.3 times a binary cross-entropy on the mask, evaluated on land cells only, because the mapped limit exists only on land. We train with the AdamW optimiser and a cosine learning-rate schedule, on $80\,\%$ of the runs, and keep the checkpoint with the lowest residual on the remaining $20\,\%$.

2.4 The inverse problem

2.4.1 Objective and fixed-point iteration updates the ELA field

We seek the LGM ELA field $E(x, y)$ such that the modelled ice extent, the mask $M_{\mathrm{mod}} = 1[h > 0]$, matches the mapped one, M_{obs}. The objective function that we try to minimise is

$$\mathrm{res} \;=\; \sum_{x,y} \big| M_{\mathrm{obs}} - M_{\mathrm{mod}} \big|, \tag{7}$$

the number of cells on which the modelled and observed presence of ice disagree.

A binary mask is used because reliable past ice thicknesses are rare. Furthermore, a mask has no usable derivative, so we cannot use a gradient method with an ice-flow model, and an inversion based on the ice extent only could not be solved with adjoint methods. Instead we follow Visnjevic and others (2018) and update the ELA from the mismatch between the modelled and the mapped extent,

$$E \;\leftarrow\; \mathcal{S}_n \Big[E - \tau \left(M_{\mathrm{obs}} - M_{\mathrm{mod}} \right) \Big], \tag{8}$$

where τ is a step length in metres and $\mathcal{S}_n$ is $n = 50$ passes of Laplacian smoothing. Where the model is missing mapped ice the ELA is lowered, and where it has ice that the map does not, the ELA is raised. The smoothing regularises the field and spreads the local mismatch over the scale of the ice response. The fixed point of the update does not depend on τ, which only sets the path towards it, but τ must be large enough for the correction to overcome the smoothing. We use $\tau = 500$ m for the SIA, where each update follows only a limited number of solver cycles, and $\tau = 200$ m for IGM and for the surrogate, where each update follows a complete run.

2.4.2 Regularisation and ill-posedness

The method we use is an instance of a classical regularisation family for ill-posed problems. The idea is not new and has been developed by Lavrentiev (1967), Tikhonov and Arsenin (1977), Morozov (1984) and, specifically for iteration procedures by Vainikko and Veretennikov (1986). It has been applied successfully for inverting ELA from observed ice extent by (Višnjević et al., 2018, 2020). To reduce the dimensionality of the problem, we follow Višnjević et al. (2018) by fixing the mass-balance gradients β_{acc} and β_{abl}, the cap c and, for the transient models, the amplitude ΔE of the forcing, and we run the inversion. We then repeat it multiple times with different values of these parameters and with different random initial fields, and treat the results as an ensemble. It is worth stressing that despite this, the problem remains ill-posed. The equilibrium-line altitude is not uniquely determined by the mapped limit, because the observation is a binary mask, so thickness, mass turnover and the seasonal distribution of the balance all lie in the null space of the observation operator. The solution also does not depend continuously on the data, the forward operator being smoothing and its inverse correspondingly unbounded. The operator $\mathcal{S}_n$ suppresses the amplification of high-frequency margin detail, while the ensemble over the gradients and the bound that the degree-day model places on them, which is explained below, address the non-uniqueness that smoothing cannot.

2.4.3 The same update with every forward model

We apply the update of Eq. (8) with every forward model, with the same smoothing and the same start, a uniform ELA plus 40 m of noise, so that only the way in which M_{mod} is obtained changes. With the SIA, each update follows 250 solver cycles of a single continuous run, in which the ice is carried over from one update to the next. With IGM, each update follows a complete transient run, and with the surrogate described below it follows one evaluation of a neural network. We use $\tau = 500$ m for the SIA and $\tau = 200$ m for IGM and the surrogate. For the latter two, we also limit the ELA to stay within ± 600 m of its starting value, which we call a clamp, to prevent the field from drifting where the extent gives no information. Because the inversion is the same, differences between the results reflect the ice model and the state of the ice that it represents, and not the inversion scheme. In practice the iteration does not always settle. The residual decreases within a few tens of iterations and then often cycles by a few hundred cells, while the ELA field changes by only a few metres, so we take the mean of the last 10 iterations for IGM and of the last 50 for the surrogate as the result.

2.4.4 A neural surrogate checked against the higher-order model and retrained during the inversion

The forward model of the update is the neural surrogate of the transient model, which replaces the higher-order IGM (Blatter–Pattyn) in the loop. Its use in the inversion is justified only by its agreement with that model, and we check every surrogate inversion against it in two ways. First, we run the same update with the higher-order IGM itself in the loop, from different starts, and compare the recovered field with the surrogate one. Second, we run forward in the higher-order IGM selected fields recovered with the surrogate, at the extremes and in the centre of the accepted set, with the same forcing, and compare the modelled extent with the surrogate prediction. A gap in cells is converted into metres of ELA with the change of extent per metre of ELA, so that it can be compared with the other uncertainties of the inversion. Where the surrogate and the higher-order IGM disagree by more than this, we add training runs around the fields concerned and retrain, and we repeat the checks. The surrogate holds for the forcing, time window, starting ice and parameter ranges of its training runs; outside them, or for another region, it is retrained.

2.4.5 Selection of ensemble members

Not all inversions of the ensemble reproduce the mapped extent equally well, and we keep those that do so within a tolerance. A member is accepted if its land residual (the mean of the last 50 iterations) is within the tolerance, which is the quadrature sum of three measured errors that reflect the scatter of the residual along the iterations (the update does not always settle, but cycles by a few hundred cells), the scatter between random starts, and the surrogate–IGM residual gap measured on one field run in IGM. We apply the selection inside the range of mass-balance gradients that the degree-day model allows, which is explained below, so that the best member is itself taken inside this range.

2.5 Numerical implementations and cost

The cost of an inversion is the number of iterations of Eq. (8) multiplied by the cost of one forward evaluation. Each iteration needs one forward run with the current ELA field, the comparison of the modelled and mapped masks, and a smoothing of the correction, which has a negligible cost. The update uses no derivative of the ice thickness with respect to the ELA. An adjoint method would require differentiating the ice-flow model and storing its time history, and it would not work with a binary mask. A derivative-free search would need at least one run per unknown, which is more than 10^4 runs for the ice-covered cells of a region, whereas the smoothing lets every cell be corrected at once from a single run. The number of iterations needed is therefore of the order of tens to hundreds.

With the SIA, the model is written in PyTorch in double precision and advanced with super-time-stepping, using $N = 6$ substeps on a Chebyshev schedule with a damping of 0.02, which allows steps far above the explicit stability limit of the diffusive flux. The mass balance is recomputed from the evolving surface at every step. In the SIA inversion, Eq. (8) is applied after every 250 cycles of a single continuous relaxation, with $\tau = 500$ m, so that the ice is carried over from one update to the next and the model is never run to equilibrium before the field is corrected. About 300 updates are needed, which takes a few minutes per field. This is what made the SIA inversion affordable, since running each iteration to a steady state takes far longer (Višnjević et al., 2018). The price is that the end state of this inversion is neither a steady nor a transient state, which is why we use it as a baseline only and not for the quantitative result.

With IGM, we use the solved method and a full transient run of 12 kyr for each iteration. One run takes one to tens of minutes on an RTX 4090 and about six times that on a TITAN X., which are the GPUs we had access to. For the example we show below, the iteration settles within 30 to 40 runs, so an inversion takes a few hours on the faster GPU. This is the reference result, but it is too slow for an ensemble, since a few hundred inversions would take thousand GPU-hours.

With the surrogate, one evaluation of the network replaces a complete IGM run, and 600 iterations take about tens of seconds. The saving comes from the combination of the two ideas. The update needs only the modelled mask and not its derivative, so a network that predicts the mask is sufficient, and it needs only tens of iterations, so the number of evaluations stays small. An ensemble of about 400 inversions, over the mass-balance gradients, the forcing amplitude and the random starts, then takes about a couple of hours, against about thousands with IGM.

2.6 From mass balance to climate

Now that we have obtained a series of ELA fields, our goal is to derive estimates of climate parameters (i.e., temperature and precipitation) at specific locations and as maps.

The mass balance we infer corresponds to accumulation ($\mathcal{A}$) minus melt ($\mathcal{M}$), each an integral over the year,

$$m = \mathcal{A}\big[P(\cdot), T(\cdot)\big] - \mathcal{M}\big[T(\cdot)\big]. \tag{9}$$

Several factors are invisible to the observable. A family of seasonal climates maps to the same annual m, hence the same h, the same M_{mod} and the same residual. The data constrain m (in practice its zero set, the ELA, and its gradients nearby) and nothing about how the year was distributed to produce it, so degree-day factors, a seasonal amplitude and a rain and snow threshold cannot be inverted for. This can only be done using a model.

Here we use a degree-day model, which takes a climate, given by an annual temperature cycle and an annual precipitation total, and returns the surface mass balance as a function of elevation. The model is evaluated month by month. The temperature of month k ($k = 1, \dots, 12$) at elevation s is

$$T_k(s) \;=\; \bar{T}_0 + A\cos\Big(\frac{2\pi(t_k - t_0)}{\tau_{\mathrm{yr}}}\Big) - \Gamma(s - s_0), \tag{10}$$

where t_k is the middle of the month, $\bar{T}_0$ is the mean annual temperature at a reference elevation s_0, Γ is the lapse rate, A is half the annual range (so the warmest month reaches $\bar{T}_0 + A$ and the coldest $\bar{T}_0 - A$), and t_0 fixes the phase.

Melt only responds to positive temperatures. A month whose mean is -2°C still contains warm days that melt ice, so using monthly means alone would predict no melt at all for that month and would systematically underestimate ablation near the ELA. We therefore treat the daily temperature as scattered about the seasonal cycle (10) as a Gaussian of standard deviation σ, and integrate the positive temperature over that distribution. Calov and Greve (2005) showed that this expectation has a closed form, so the positive degree days of month k are

$$\mathrm{PDD}_k(s) = \frac{1}{12}\Big[\underbrace{\frac{\sigma}{\sqrt{2\pi}}e^{-T_k^2/2\sigma^2}}_{\text{warm days when the mean is cold}} + \underbrace{\frac{T_k}{2}\operatorname{erfc}\Big(\frac{-T_k}{\sigma\sqrt{2}}\Big)}_{\to\, T_k \text{ once } T_k \gg \sigma}\Big], \tag{11}$$

in units of K yr. No stochastic simulation is required and the two terms recover the expected behaviour in both limits, tending to zero for $T_k \ll -\sigma$ and to T_k itself for $T_k \gg \sigma$.

Accumulation is the precipitation that falls as snow rather than rain. We take the annual precipitation P (m w.e. per year) to be spread evenly over the year and constant with elevation, so that the snowfall of month k is $a_k = \varphi(T_k)\,P/12$, where the snow fraction φ ramps linearly from all snow below 0°C to all rain above 2°C,

$$\varphi(T) = \mathrm{clip}\Big(\frac{2 - T}{2}, 0, 1\Big), \qquad \mathcal{A}(s) = \sum_{k=1}^{12} a_k. \tag{12}$$

Melt is given by the positive degree days multiplied by a degree-day factor. Two factors are needed because the surface changes through the melt season. Fresh snow is bright and melts slowly, while the bare ice exposed once the snowpack has gone is darker, absorbs more radiation and melts roughly 50 to 100% faster. The snowpack is carried from month to month. In month k the available snow is melted first at $f_{\mathrm{snow}}^{\mathrm{DDF}}$, and only the positive degree days left over after the snow is exhausted melt

the underlying ice at the larger $f_{\mathrm{ice}}^{\mathrm{DDF}}$,

$$m_k^{\mathrm{snow}} = \min\bigl(S_{k-1} + a_k,\, f_{\mathrm{snow}}^{\mathrm{DDF}}\mathrm{PDD}_k\bigr), \qquad S_k = S_{k-1} + a_k - m_k^{\mathrm{snow}}, \tag{13}$$

$$\mathcal{M}(s) = \sum_{k=1}^{12}\Bigl[m_k^{\mathrm{snow}} + f_{\mathrm{ice}}^{\mathrm{DDF}}\Bigl(\mathrm{PDD}_k - \frac{m_k^{\mathrm{snow}}}{f_{\mathrm{snow}}^{\mathrm{DDF}}}\Bigr)\Bigr], \tag{14}$$

where S_k is the snowpack at the end of month k. A fraction $r = 0.6$ of the melt refreezes in the snowpack. The annual balance is then accumulation minus the melt that is not refrozen,

$$m(s) \;=\; \mathcal{A}(s) \;-\; (1-r)\,\mathcal{M}(s), \tag{15}$$

and the ELA is where $m = 0$.

At any given location, the model needs a present-day temperature, which we calculate. Fixing A, σ, P and Γ leaves the mean annual sea-level temperature $\bar{T}_0$ as the only free parameter, so requiring the model to reproduce the observed modern ELA at that site is one equation in one unknown. The solution is the present-day climate against which the glacial perturbation is measured.

With that model we can evaluate the modern and the past position of the ELA. Our objective, however, is the change in climate that separates them. We therefore ask which perturbation of the modern climate reproduces the glacial ELA, which is inferred from the inversion,

$$\mathrm{ELA}\bigl(\;\underbrace{\bar{T}_0 + \Delta T}_{\text{temperature change}}\;,\;\underbrace{P(1+\Delta P)}_{\text{precipitation change}}\;\bigr) \;=\; \mathrm{ELA}_{\mathrm{LGM}}. \tag{16}$$

Eq. (16) is a single equation, since the modelled ELA must equal one number, but it contains two unknowns: the difference in temperature, ΔT, and precipitation, ΔP. The ice extent therefore cannot give both. We fix ΔP at a chosen value, solve for ΔT, and repeat over a plausible range of ΔP. This traces a curve of admissible $(\Delta P, \Delta T)$ pairs and that curve, which call ΔP-Δ curve, simply reflects that a wetter glacial climate needs more cooling to reach the same ELA, and a drier one needs less. Near the modern climate the curve is close to a straight line, which follows from linearising Eq. (16),

$$\frac{\partial \mathrm{ELA}}{\partial T}\,\Delta T + \frac{\partial \mathrm{ELA}}{\partial P}\,\Delta P \;=\; \mathrm{ELA}_{\mathrm{LGM}} - \mathrm{ELA}_{\mathrm{mod}}, \tag{17}$$

where the two sensitivities are obtained by perturbing the forward model. Here $\partial\mathrm{ELA}/\partial T > 0$ (a warmer climate raises the ELA), $\partial\mathrm{ELA}/\partial P < 0$ (more precipitation lowers it), and the right-hand side is negative. The ratio $-(\partial\mathrm{ELA}/\partial P)/(\partial\mathrm{ELA}/\partial T)$ is the exchange rate along the curve and sets how much extra cooling is needed to compensate a given drying. At unchanged precipitation, $\Delta P = 0$, Eq. (17) gives $\Delta T = \Delta\mathrm{ELA}\,/\,(\partial\mathrm{ELA}/\partial T)$, which is simply $\Gamma\,\Delta\mathrm{ELA}$, since $\partial\mathrm{ELA}/\partial T = 1/\Gamma$.

Picking a point on this curve needs information that the ice extent does not carry. One option is an independent temperature estimate, from climate proxies (Bartlein et al., 2011; Prentice et al., 2017) or luminescence thermometry (Biswas et al., 2020).

Another is a physical argument. A colder atmosphere holds less water vapour, by about 7% per kelvin (Clausius–Clapeyron), and if precipitation scales with it, then

$$1 + \Delta P \;=\; \exp(\alpha\,\Delta T), \qquad \text{so} \qquad \Delta P \simeq \alpha\,\Delta T, \tag{18}$$

with $\alpha = 0.07\ \mathrm{K}^{-1}$ as the central value and 0.05 to $0.09\ \mathrm{K}^{-1}$ explored. Both ΔP and ΔT are negative here, meaning that cooling brings drying. Eq. (16) and Eq. (18) are two curves in the $(\Delta P, \Delta T)$ plane, and their intersection is one admissible point. Each member of the ensemble of accepted inversion results gives its own ΔP-Δ curve and hence its own intersection. We report the median of these points and the 5 to 95 percent range of the combined uncertainty described below.

It is important to stress that the Clausius–Clapeyron relation captures only the thermodynamic change of atmospheric moisture with temperature. Changes in atmospheric circulation, such as shifts of the storm tracks or the jet streams, and changes in the interaction of the flow with the orography of mountain ranges, can modify precipitation substantially and are not included. The Clausius–Clapeyron relation should therefore be read as a reference that selects one point on the trade-off between cooling and precipitation. The uncertainty that we propagate covers the range of the Clausius–Clapeyron scaling, but it does not include these dynamical changes.

Seasonality adds a third unknown. So far the perturbation has shifted the whole annual cycle uniformly. A climate can also change the contrast between summer and winter while leaving the annual mean alone. Allowing the half-range in (10) to change, $A \to A + \Delta A$, the summer and winter changes are no longer equal,

$$\Delta T_{\mathrm{sum}} = \Delta T_{\mathrm{ann}} + \Delta A, \qquad \Delta T_{\mathrm{win}} = \Delta T_{\mathrm{ann}} - \Delta A. \tag{19}$$

This matters because melting happens almost entirely in summer. The positive degree days are negligible in months whose temperature stays well below freezing, so winter conditions barely affect the annual balance. The ELA is therefore sensitive to ΔT_{sum} and almost blind to ΔT_{win}. The consequence is that the ice determines the summer cooling reasonably well, but yields the annual mean only once ΔA is assumed. We sample ΔA between -1 and $+1$ K, and we also scale the modern half-range of each site between 0.7 and 1.3 times its assumed value.

Below, we present maps of the ELA, temperature–precipitation relationships at chosen locations, and inferred temperature-change maps under selected precipitation scenarios.

2.6.1 Gradients implied by the degree-day model

A bound exists on the mass-balance gradients which the ice extent inversion cannot supply. The forward model parameterises the mass balance by the linear law (2), whose gradients β_{abl} and β_{acc} we treat as free and determine by sweeping. Neither is bounded by the residual alone. The degree-day model supplies a constraint, because it predicts what those gradients should be rather than fitting them. Differentiating the degree-day balance with respect to elevation at the ELA reproduces the quantities

that (2) parameterises,

$$\beta_{\mathrm{abl}} \simeq f_{\mathrm{ice}}^{\mathrm{DDF}} \left.\frac{\mathrm{dPDD}}{\mathrm{d}s}\right|_{\mathrm{ELA}}, \qquad \beta_{\mathrm{acc}} \simeq \left.\frac{\mathrm{d}\mathcal{A}}{\mathrm{d}s}\right|_{\mathrm{ELA}}. \tag{20}$$

The left-hand sides are free parameters of the ice-flow model, and the right-hand sides are determined by melt factors and climate. Equating them converts measurements made on modern glaciers into a bound on parameters the glacial ice extent leaves open. One refinement is needed in practice. Eq. (2) assumes a constant gradient throughout the ablation zone, whereas the degree-day balance steepens with depth below the ELA, because lower ground is warmer for longer each year. The appropriate comparison is therefore the best linear fit to the degree-day balance over the depth of ablation zone the model actually contains.

2.6.2 Temperature and precipitation changes at key locations and regionally

As we show above, the conversion of the ELA into temperature and precipitation changes needs three climate inputs, which are the modern ELA, the modern precipitation and the half-range of the seasonal temperature cycle. We therefore apply it first at a small number of key locations and then extend it to a map. The key locations are sites where the modern ELA is known from a monitored glacier, for example from long-term snowline or mass-balance records, and where plausible values of the annual precipitation and of the half-range of the seasonal temperature cycle can be assigned.

To obtain maps, the modern ELA is needed everywhere on the mapped ice. For this we use all the monitored glaciers of the region, and we fit a smooth, low-order surface through their reference ELAs, and we record its residual standard deviation as a measure of the error. The precipitation and the half-range are only available at the key locations, so between them we interpolate them as functions of the local modern ELA. This is an assumption, and it is the least constrained part of the conversion. The conversion is then applied cell by cell, with the local modern ELA and the local ELA of the inversion, and gives maps of cooling and of precipitation change. As with distance the extrapolation becomes unreliable, so we exclude cells beyond a chosen distance from the statistics. The same conversion is applied to each member of the ensemble of ELA fields, and the spread between members is part of the uncertainty of the map.

2.6.3 Precipitation that is constant with elevation

In the degree-day model we assume the precipitation rate P to be constant with elevation, so accumulation varies with height only through the snow fraction $\varphi(T)$. In mountains precipitation typically increases with elevation up to some level and decreases above it, and it also varies strongly across drainage divides. With constant P, cooling by ΔT is equivalent to moving up by $\Delta T/\Gamma$, so that $\partial \mathrm{ELA}/\partial T = 1/\Gamma$ and $\Delta T = \Gamma\,\Delta\mathrm{ELA}$ at unchanged precipitation, and these relations hold exactly only for elevation-independent P. We use a constant P as the reference, because the profile at each site is often not known from the data, and we include the effect of an elevation profile in the uncertainty.

2.6.4 Uncertainty propagation

We propagate the uncertainty of the cooling and of the precipitation change using a Monte Carlo approach, drawing all sources together from ranges that we consider plausible. Some sources come from the inversion. These are the choice of member

among the accepted ones, and the bias of the surrogate ELA relative to an inversion run with the full ice model, which we sample around the value measured. Others come from the climate step. These are the lapse rate (5 to $7\,\mathrm{K\,km^{-1}}$), the change in the half-range of the seasonal cycle ($\pm 1\,\mathrm{K}$) and its scale (0.7 to 1.3), the precipitation (0.5 to 1.5 times its assumed value), the Clausius–Clapeyron constant (0.05 to $0.09\ \mathrm{K^{-1}}$) and the elevation dependence of the precipitation. For the last one we use a profile that rises with elevation up to a fixed level and falls at the same rate above it, referenced to the precipitation at the modern ELA, with a rate drawn between zero and $40\,\%$ per kilometre. The last source is the error of the modern ELA, which is the error assigned at the key locations and the residual of the fitted surface for the maps. To rank the sources, we also vary each one alone.

The lapse rate is varied, but it is held equal today and at the LGM, so that ΔT is the shift of the whole temperature profile and is quoted at the ELA. A glacial lapse rate that differs from the modern one would change the inferred cooling, since $\Delta T_{\mathrm{sl}} = \Gamma_{\mathrm{LGM}} z_{\mathrm{LGM}} - \Gamma_{\mathrm{mod}} z_{\mathrm{mod}}$, and the ice extent cannot test this assumption.

We report two kinds of ranges. The spread between the accepted members, which we show on the maps, is the range between the 10th and the 90th percentile of the members at each cell. The accepted ensemble has few members, and this range, which contains most of them, is less sensitive to a single member than the 5 to $95\,\%$ range. The total uncertainty, which comes from the Monte Carlo step with many draws, is given as the 5 to $95\,\%$ range.

3 Example: the South Island of New Zealand

We apply the method to the South Island of New Zealand in five steps. First, we describe the data used in the inversion. Second, we perform an initial inversion using the computationally inexpensive shallow-ice approximation (SIA) to obtain a first estimate of ELA fields. These inferred fields guide the initial runs of IGM, which solves the higher-order Blatter–Pattyn equations and provides the training data for the surrogate model. Third, we use the trained surrogate to perform inversions over a range of mass-balance gradients. We compare the results with those obtained using the SIA and with a limited number of inversions performed directly with IGM, allowing us to distinguish the effects of the ice-flow model from those of the ice state. We then use the inversion results to generate additional training data, retrain the surrogate, and repeat the inversion until satisfactory convergence is achieved. Fourth, we analyse the inferred climate using a positive degree-day (PDD) model and the Clausius–Clapeyron relationship. Finally, we assess the sensitivity of the recovered ELA to the forcing, initial ice geometry, and spatial resolution, and translate the inferred ELA changes into corresponding temperature and precipitation changes.

3.1 Data and previous model

The South Island has a well-dated moraine record (Barrell et al., 2011) and a strong precipitation gradient across the Southern Alps, which was modelled at the LGM by Golledge et al. (2012) with a PDD ice-sheet model forced by a uniform cooling. We use the mapped limit on a 399×366 grid at 2 km ($15{,}095$ ice cells), a cap $c = 3\,\mathrm{m\,a^{-1}}$, and a 41-glacier set from the end-of-summer-snowline programme to define the modern ELA (a plane of $+973\,\mathrm{m}$ per 100 km across the strike, residual $146\,\mathrm{m}$). The glacial index s is based on the EPICA Dome C oxygen data; over -30 to $-18\,\mathrm{ka}$ it varies only between 0.85 and 1.08 (Fig. 1).

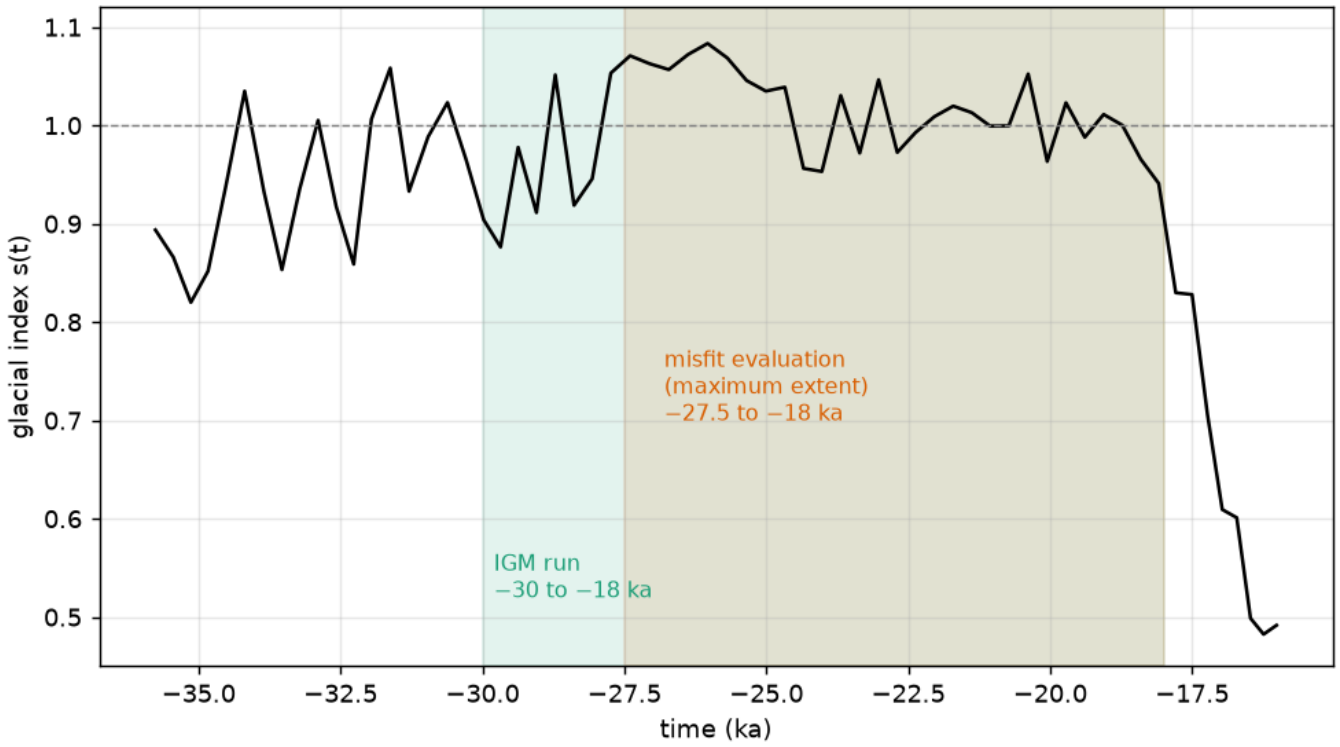


Figure 1. Glacial index $s(t)$ used to force the model, from the EPICA Dome C ice-core record. The index is 0 for the Holocene (mean of 0 to 2 ka) and 1 for the mean of 18 to 24 ka, and it scales the change of the ELA with time (Eq. 5). The green band is the period simulated in each IGM run, from -30 to -18 ka. The orange band is the part of the run over which the maximum modelled extent is compared with the mapped limit to evaluate the misfit, from -27.5 to -18 ka. The first 2.5 kyr of each run is not used in the misfit. They only set the initial condition. Over this period the index varies little, between 0.85 and 1.08.

We use melt factors ($f_{\mathrm{snow}}^{\mathrm{DDF}} = 4.55$ and $f_{\mathrm{ice}}^{\mathrm{DDF}} = 7.17\,\mathrm{mm\,w.e.\,K^{-1}\,d^{-1}}$) and the daily scatter ($\sigma = 2.5°\mathrm{C}$) from Anderson et al. (2006), as used by Golledge et al. (2012). Precipitation falls as snow below 0°C and as rain above 2°C, with a linear ramp between the two, and 60 % of the melt refreezes in the snowpack. The central lapse rate is $\Gamma = 6°\mathrm{C\,km^{-1}}$, and values between 5 and $7°\mathrm{C\,km^{-1}}$ are explored below.

We selected three glaciers, at Mt Irene, Tasman and South Cameron, with modern ELAs of 1,563, 1,790 and 2,250 m taken from the long-term reference values of the end-of-summer snowline survey (Lorrey et al., 2022), to which we assign an error of ±50 m. The precipitation P and the half-range A of the seasonal cycle at each site are assumed and not measured, with $P = 6$, 4 and $2\,\mathrm{m\,w.e.\,a^{-1}}$ and $A = 3.5$, 4.0 and 5.0 K. With this climate and $\Gamma = 6°\mathrm{C\,km^{-1}}$, the degree-day model gives effective gradients $\beta_{\mathrm{abl}} = 0.0087$, 0.0081 and 0.0052 and $\beta_{\mathrm{acc}} = 0.0068$, 0.0048 and 0.0029 at the modern ELA, which vary by 40 to 70 % between the wet west and the dry east, whereas the ice model uses a single pair for the whole domain. As we show below, the best-fitting inversions favour a steeper ablation gradient than these values, up to 0.016, the edge of the explored range. We therefore report the results for all accepted members and for the subset whose gradients lie in the range implied by the degree-day model (β_{abl} from 0.004 to 0.009 and β_{acc} from 0.0025 to 0.006).

For the maps, we fit a plane through the measured ELAs of 41 glaciers of the survey. The plane rises by about 973 m per 100 km across the strike of the Southern Alps and has a residual standard deviation of 146 m, which corresponds to about 0.9 K at unchanged precipitation. Beyond 60 km from the nearest of these glaciers the plane is an extrapolation, and the cells are masked.

3.2 SIA inversion

We start by reporting the inversion using the SIA accelerated scheme. In Fig. 3, we show the inversion result for $\beta_{\mathrm{abl}}/\beta_{\mathrm{acc}} = 0.010/0.005$ and find the SIA reproduces the map to 2,606 of 15,095 cells (17 %), mostly at the margin; the ELA is low in the wet southwest and high in the dry east. The fit reproduces the expected W-E gradients (Barrell et al., 2011; Golledge et al., 2012; Višnjević et al., 2018),the residual falls toward the steepest ablation and gentlest accumulation gradient of the grid, and the median ELA moves by 330 m across it.

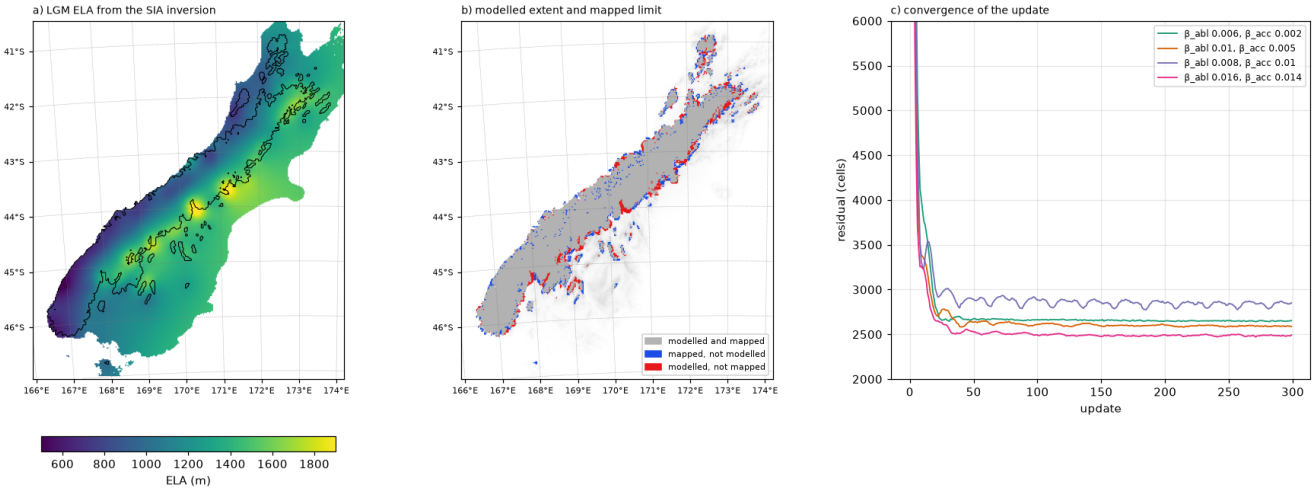


Figure 2. SIA inversion ($\beta_{\mathrm{abl}}/\beta_{\mathrm{acc}} = 0.010/0.005$). (a) ELA; (b) extent against the map; (c) the inversion converges in 300 updates; (d) relaxing the SIA fields at constant ELA for 3 kyr; (e) new ice after relaxation of the $0.008/0.010$ field.

We then run a sweep over $(\beta_{\mathrm{abl}}, \beta_{\mathrm{acc}})$ to produce an SIA ensemble of 30 fields. We find that the end state is not an equilibrium in most cases. The nudge scheme we use in the inversion converges in 300 updates (Fig. 3c), but each update only runs 250 solver cycles, which far less than the ice response time (thousands of years). Held at constant ELA, the recovered fields continue to grow $+0.2$, $+21$, $+42$ and $+41$ % in extent after 3 kyr for four members of the ensemble, the inversion converged to and the volume of the $0.010/0.005$ field by 67 % (Fig. 3). The SIA field is neither a steady state nor a transient. Nevertheless, we use these results to generate data to train the surrogate. We stress here that this inversion result enables us to significantly reduce the number of IGM, i.e., Blatter—Pattyn solutions, runs.

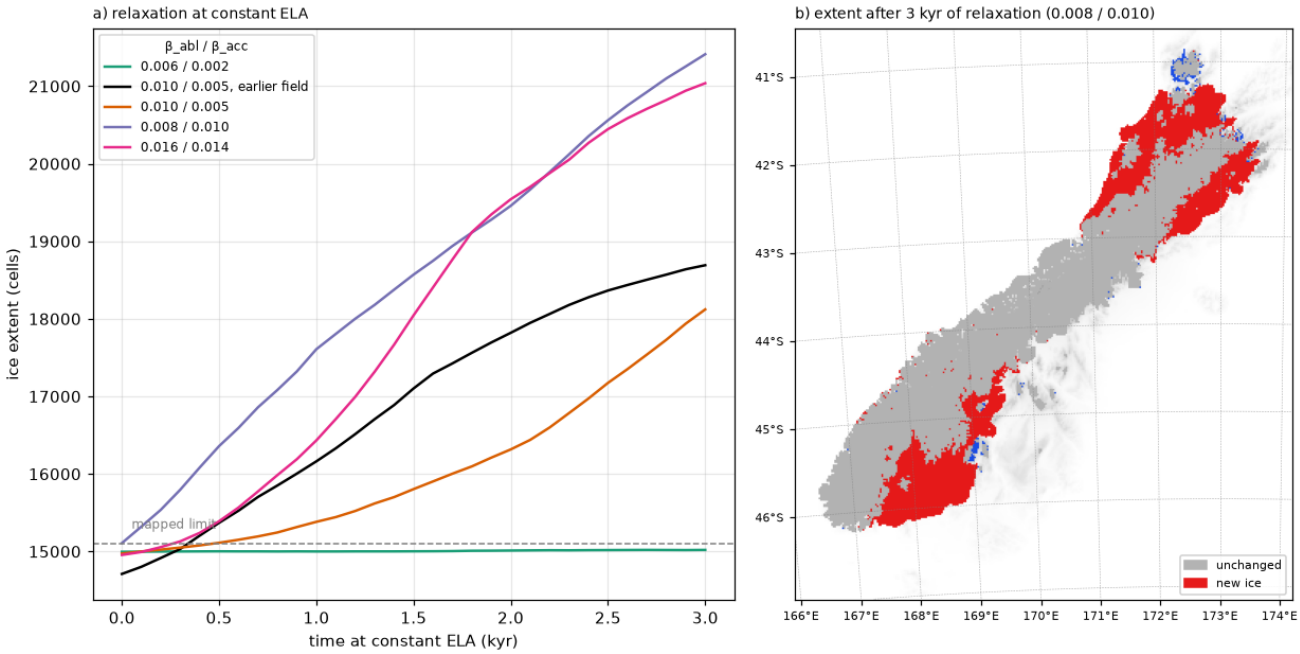


Figure 3. Relaxation of the SIA inversions at constant ELA. (a) Ice extent against time at constant ELA, for fields obtained with four pairs of mass-balance gradients (the earlier field for the pair 0.010/0.005 is shown in black). The dashed line is the mapped limit. (b) Extent after 3 kyr of relaxation for the field with $\beta_{\text{abl}} = 0.008$ and $\beta_{\text{acc}} = 0.010$. Grey is ice that was present at the start and red is new ice. The end state of the inversion is not a steady state, and the ice keeps growing.

3.3 IGM data generation and surrogate training

As mentioned above, we generate the training data progressively. We first run IGM for 1,201 ELA fields obtained by perturbing the SIA-inferred ELA field. In Fig 4, we show examples of IGM runs that used to train the surrogate.

A surrogate trained on these runs allowed us to perform the inversion rapidly, but some of the inferred ELA fields may move during the inversion process into a region of parameter space far from the SIA field, where the surrogate was not trained. We therefore run IGM for a second set of 369 fields, obtained by perturbing fields that already reproduce the mapped ice extent. Retraining on both sets improves the fit, but the surrogate still underestimated the ice on the South Island foreland. For the third set, we therefore introduce an ELA patch in that specific region and similarly along the margin of the mapped ice, generating an additional 482 usable runs out of 800 attempted. The 2,052 runs from all three sets are then used for the final training. The final surrogate inversion has a residual of 800 cells (Fig. 5). More importantly, we observe a significant reduction of the residual decreases from 1,841 to 800 between the first and last inversion. Thus, it is the additional runs targeted to the region explored by the inversion that provide the largest improvement in surrogate accuracy.

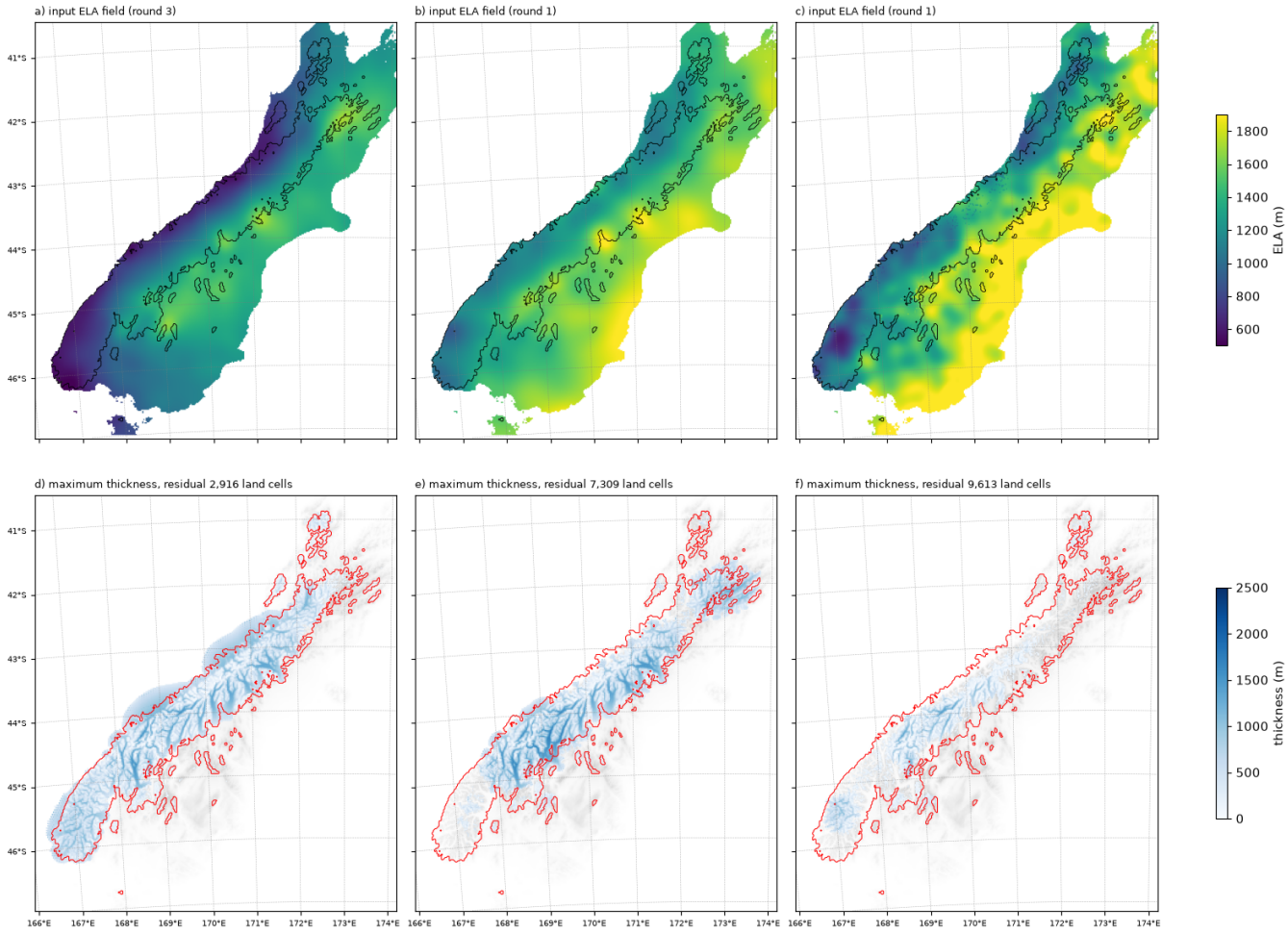


Figure 4. Three examples of IGM runs used to train the surrogate. (a to c) Input ELA field, with the mapped limit in black. (d to f) Maximum thickness over period over which the model is evaluate agains the data, with the mapped limit in red and the residual, in land cells, between the modelled and mapped extent. The runs show a good fit (d), a partial fit (e) and a poor fit (f), which the surrogate must reproduce to be used in the inversion.



Figure 5. Training data of the surrogate. (a) Outcome of the IGM runs in each of the three rounds. Usable runs are kept, and runs that put ice over the sea or flood the domain are aborted by the guard. (b) Coverage of the usable runs in the plane of the mean ELA over the mapped ice and the across-strike gradient of the ELA field. The first round, which perturbs the SIA field (black square), does not cover the region where the inversions end (IGM, star, and surrogate inversions with and without the bound on the mass-balance gradients, circles). The second and third rounds fill this region.

3.4 Inversion through the surrogate

The first step of the inversion consists in tuning the update of Eq. (8). We find that with $\tau = 20$ m the smoothing dominates and the residual never improves, and with $\tau = 500$ m it oscillates. With $\tau = 200$ m, 50 smoothing passes and a clamp of ± 600 m, the residual falls within a few tens of iterations and then cycles by about 200 cells around a stable field, whose ELA changes by only a few metres (Fig. 6). The result is the mean of the last 50 iterations. Starting from the SIA field or from a uniform ELA gives the same residual, so the answer does not depend on the initial condition.

A good fit to the mapped extent does not show that the ELA is right, because the surrogate could be wrong where we use it. We therefore run the same update in IGM, i.e., Blatter-Pattyn, with a full run at each iteration, from a uniform start and from the SIA field. Both end at the same field (rms difference 10 m over the ice, correlation 0.999), with a residual of 3,151 and 3,055 cells (all cells) that cycles by a few hundred cells. The surrogate field agrees with them over the ice, with a correlation of 0.99, an rms difference of 31 m and a bias of -3 m for the unbounded set (Fig. 6c). We also run a representative surrogate field forward in IGM. Its land residual is 2,840 cells in IGM against 2,772 predicted by the surrogate, and for a field inside the bounded range of gradients 3,058 against 2,931. The surrogate therefore reproduces the IGM inversion and the IGM forward run to within an acceptable tolerance, which we use below.

With the check passed, we can preform the inversion with the surrogate.. We show an example of how the ELA field improves during the iterative process for a single inversion in Fig. 7.

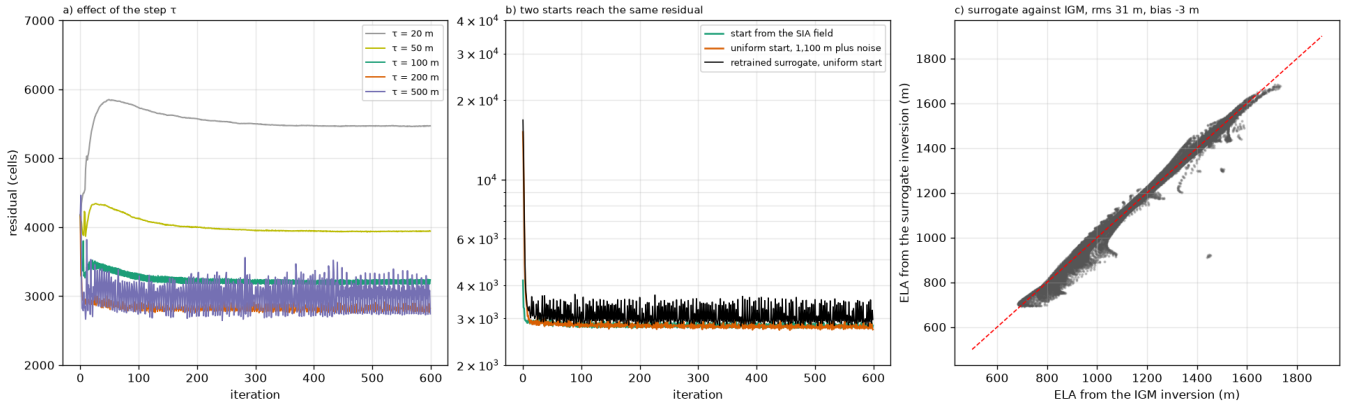


Figure 6. Inversion through the surrogate. (a) Residual of the extent against the iteration for five values of the step τ. Too small a step is swamped by the smoothing and too large a step oscillates, and $\tau = 200$ m converges and is stable. (b) Two different starts, the SIA field and a uniform ELA plus noise, reach the same residual, and so does the retrained surrogate. (c) ELA at every cell of the mapped ice from the surrogate inversion, as the median of the accepted members, against the ELA from the inversion run in IGM itself.

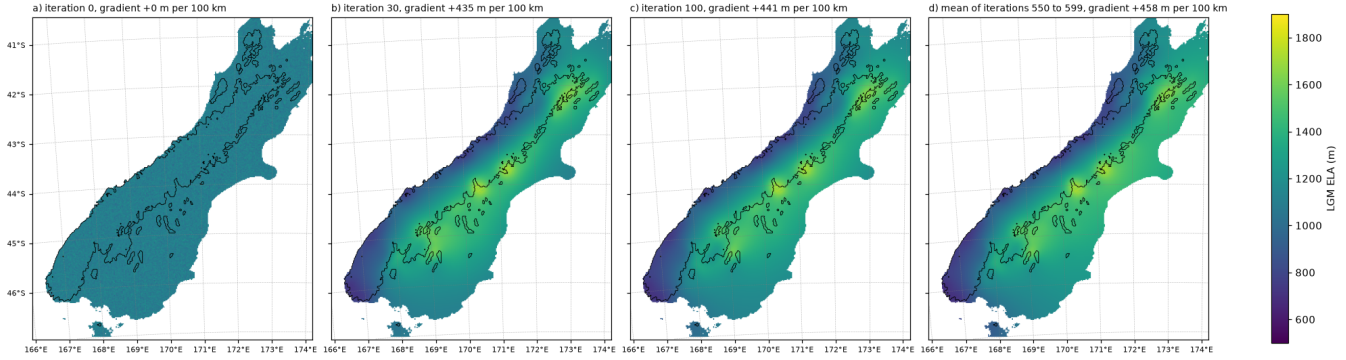


Figure 7. Evolution of the ELA field during the inversion through the surrogate, from a uniform start. The mapped limit is shown in black. The field acquires the across-strike gradient within 30 iterations and changes very little afterwards. The last panel is the mean of iterations 550 to 599.

We then run the inversion over an 11×9 grid in $(\beta_{\mathrm{abl}}, \beta_{\mathrm{acc}})$, with amplitudes of 500, 700 and 900 m and two random starts, which is 396 inversions in about 1 h. The residual decreases towards steep ablation gradients and gentle accumulation gradients with no interior minimum, while the median ELA over the mapped ice spans 994 to 1,329 m across the grid (Fig. 8). We keep the members whose land residual and intersection over union are within a tolerance of the best, which is the quadrature sum of the cycle scatter, the scatter between starts and the surrogate–IGM gap (538 cells, 20 % of the best residual). This accepts 165 members, with β_{abl} from 0.006 to 0.016 and β_{acc} up to 0.007.

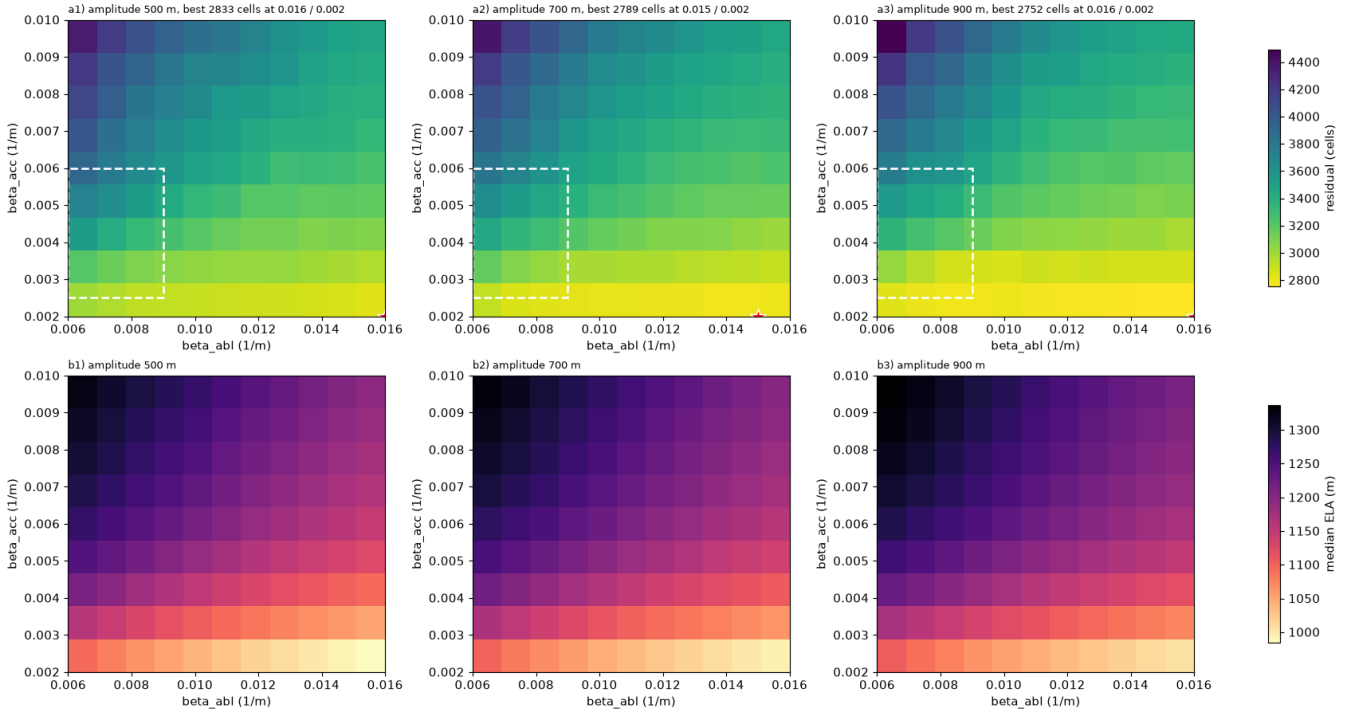


Figure 8. Surrogate inversion over the mass-balance gradients for three forcing amplitudes, South Island of New Zealand. Top row, extent residual for each pair of the ablation gradient β_{abl} and the accumulation gradient β_{acc}. Bottom row, median recovered ELA over the mapped ice. The columns are for amplitudes of 500, 700 and 900 m, and the colour scales are the same in all columns. The star marks the pair with the lowest residual in each column, and the dashed box marks the range of gradients allowed by the degree-day model (the grid starts at $\beta_{\mathrm{abl}} = 0.006$, so the box is truncated on its left). Each cell is an individual inversion, and for the amplitude of 700 m it is the mean of two inversions with different random starts. The residual varies by less than 2,000 cells over the grid and by less than 3 % between amplitudes, and the median ELA spans about 335 m across the grid (985 to 1,322 m for 500 m and 1,006 to 1,337 m for 900 m), whereas the amplitude shifts it by about 20 m.

The accepted inversion results still include gradients that the degree-day model cannot produce, with ablation gradients above 0.009. We therefore repeat the selection inside the range of gradients that the degree-day model allows (Eq. (20), $0.004 \leq \beta_{\mathrm{abl}} \leq 0.009$ and $0.0025 \leq \beta_{\mathrm{acc}} \leq 0.006$). Of the 396 members, 48 lie in this range, 20 pass the acceptance rule, and the best residual inside the range is only 4 % larger than over the whole grid. The data do not reject the physically consistent gradients, and the spread of the ELA between the members is more than halved. We use these 20 members as the main result and the 165 as the alternative.

3.5 Comparison of the SIA, surrogate and IGM inversions

The transient IGM (Blatter–Pattyn and surrogate) inversions lead to an ELA field that differs from the SIA one (Fig. 9a–d). Along the across-strike direction (towards the southeast, perpendicular to the Southern Alps) the ELA gradient is 549 m per 100 km in the ensemble of SIA inversions, 457 m per 100 km with the surrogate (the median of the 165 accepted members, which range from 444 to 471), and 423 and 436 m per 100 km with the inversions run in IGM from a uniform start and from the SIA field, against 973 m per 100 km for the modern ELA. Restricting the surrogate ensemble to the gradients allowed by the degree-day model (20 members) gives 454 m per 100 km. The mean ELA changes little between the inversions (1,136 m for the SIA and IGM and 1,133 m for the surrogate), but it is the least constrained property. It ranges from 991 to 1,223 m between the surrogate members, and it is 1,179 m for the members inside the degree-day range. The main difference between the SIA and the other inversions is the rise of the ELA from west to east.

The two inversions run in IGM end at the same field, with a root-mean-square difference of 10 m over the ice and a correlation of 0.999, so the result does not depend on the start. The surrogate agrees with them. Its median field differs from the IGM field by 31 m (root mean square) with a bias of -3 m. As a further check, we ran a representative member of each surrogate ensemble forward in IGM. The land residual is 2,840 cells in IGM against 2,772 predicted by the surrogate for the full ensemble, and 3,058 against 2,931 for the members inside the degree-day range.

Including the longitudinal stresses in IGM does not explain the difference with the SIA. Running the same ELA field through the SIA and through IGM with the same transient forcing gives the same land extent (20,537 against 20,268 cells, within 5 % in every across-strike band, Fig. 9e), and IGM only adds ice over the sea, which the SIA does not have. The difference comes from the state of the ice in the SIA fields. The SIA field is not a transient solution. The nudge scheme runs only 250 solver cycles per update, so the ice and the ELA evolve together, and at a constant ELA the SIA ice would grow by a further 27 % in extent after 3 kyr, as reported above. A transient inversion, in which the ice has a finite, specified time to grow, requires a higher ELA in the west and a lower one in the east, hence the weaker gradient.

The recovered field is insensitive to the forcing (Table 1) and to the time the ice is given to grow. Starting from the converged field, changing the amplitude from 900 to 500 m, moving the index peak by ± 2 kyr, or removing the variability altogether changes the gradient by at most 4 m per 100 km and the cooling at the sites by at most 0.3 K. Starting the window 0.5, 2.5, 8.5 and 14.5 kyr before the period over which the model is evaluated gives gradients of 436 to 437 m per 100 km, a mean ELA of 1,127 to 1,149 m, and fields that differ from the baseline by 11 to 29 m (root mean square). Both results are properties of the EPICA index over this window, which varies little (it stays between 0.91 and 1.06 of its glacial mean), and they should not be assumed elsewhere, for example in the northern-hemisphere regions where the index varies much more.

The extent is also insensitive to the initial ice. For the converged field, starting from no ice at -42 or -60 ka, or from half of the SIA ice, gives a maximum land extent of 60,676, 60,664 and 61,020 km^2, against 62,400 km^2 from the SIA ice, which is less than 1 m of ELA among the first three and 3 to 4 m between them and the SIA ice (about 500 km^2 per metre of ELA). On a 1 km grid the same field gives 59,429 km^2, 4.7 % less than at 2 km and equivalent to about 6 m of ELA. This comparison is approximate, because the starting ice is interpolated and about 10 % larger and the mapped limit is rasterised differently on the

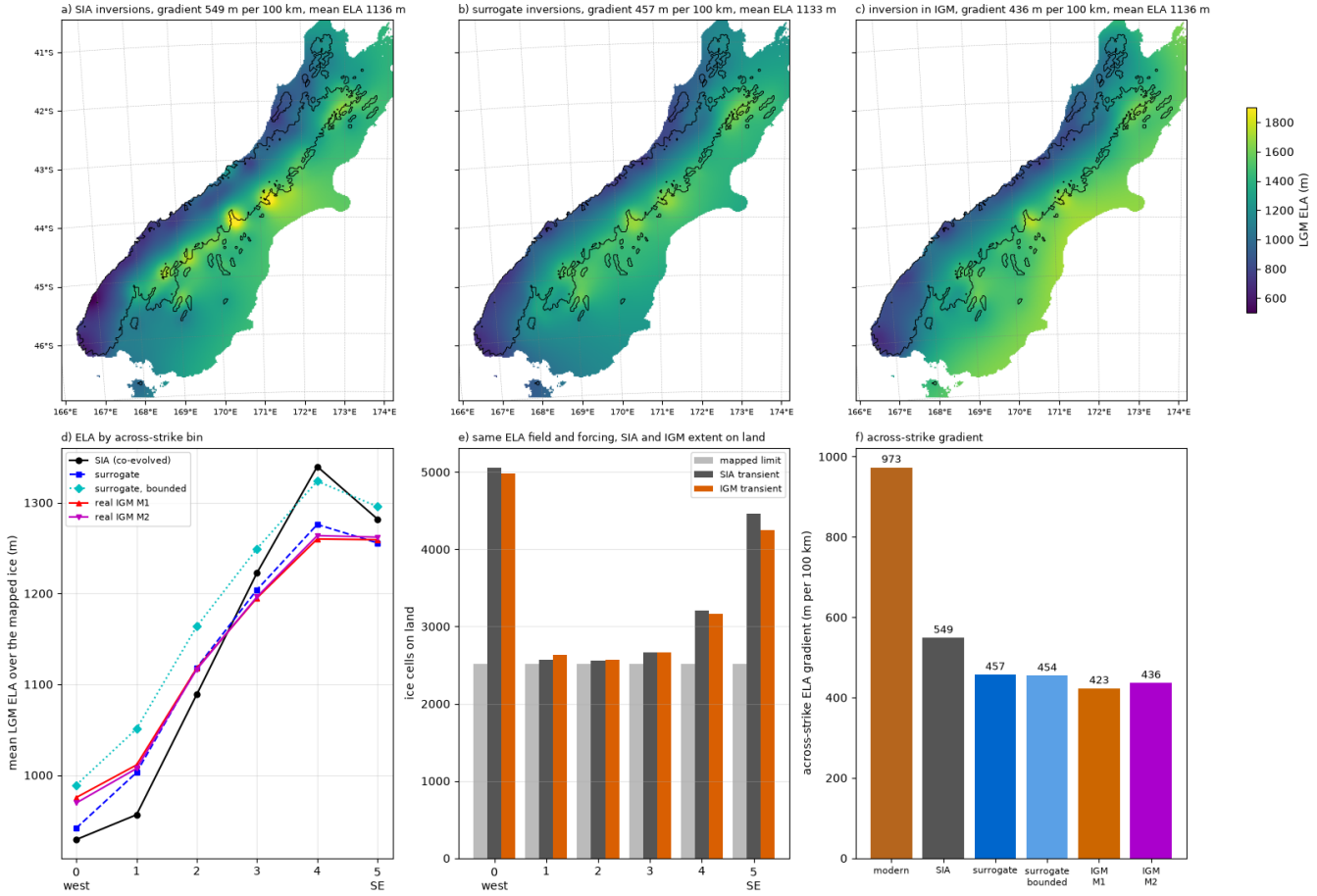


Figure 9. Comparison of the ELA from the inversions. (a to c) ELA from the SIA inversions (median of the ensemble), the surrogate inversions (median of the accepted members) and the inversion run in IGM, with the mapped limit in black. (d) Mean ELA over the mapped ice in six bins across the strike, from the west (0) to the southeast (5). (e) Land extent of the same ELA field run through the SIA and through IGM with the same forcing, by bin. (f) Across-strike gradient of the ELA.

IGM inversion, from the converged field	gradient (m/100 km)	mean ELA (m)	ΔT at the CC point (K)	
			Mt Irene / Tasman	South Cameron
baseline (EPICA, amp 900)	436	1,136	5.4 / 5.4	7.3
amp 500	436	1,116	5.4 / 5.6	7.4
peak 2 kyr earlier	439	1,149	5.1 / 5.4	7.3
peak 2 kyr later	436	1,134	5.4 / 5.4	7.3
constant ELA	440	1,101	5.4 / 5.7	7.6

Table 1. Sensitivity of the IGM inversion to the forcing.

two grids. We did not test higher resolution than 1 km, which is important to reproduce the observed ice thickness in the Alps (Leger et al., 2025).

We also ran the final field forward in IGM to a steady state (Fig. 10). Shifted uniformly by δ, the field was run with a constant ELA for 60 kyr (steady state), with a constant ELA for 12 kyr, and with the transient forcing used in the inversion (EPICA, amplitude 700 m, 12 kyr). The transient run gives 15,429 cells at $\delta = 0$ (15,095 mapped), the constant ELA 14,178 cells, and the steady runs are smaller at every shift, 13,037 to 8,366 cells for $\delta = +10$ to $+60$ m. For $\delta \leq 0$ no steady state exists: the ice fills the domain. The converged branch, extrapolated to $\delta = 0$, reaches about 13,800 cells, so it ends (a fold) at about 14,000 cells, some 8 % below the mapped extent. The slopes are of the same order in the three cases (80, 63 and 71 cells per metre of ELA for the steady state, the 12 kyr constant ELA and the transient run between $+30$ and $+60$ m), so at the same extent a steady-state inversion needs an ELA 25 to 28 m lower than the transient one. The steady state holds 72 % (west) and 71 % (southeast) of the transient extent at $+30$ m, against 95 % in the middle of the range (Fig. 10b). Two features explain the sign. The transient forcing is slightly colder than the constant ELA at the peak (the index reaches 1.06, so the ELA is lowered by up to about 40 m), and the steady runs are still losing the starting ice. The test shifts one field uniformly, so the fold is a property of this field and not of the data.

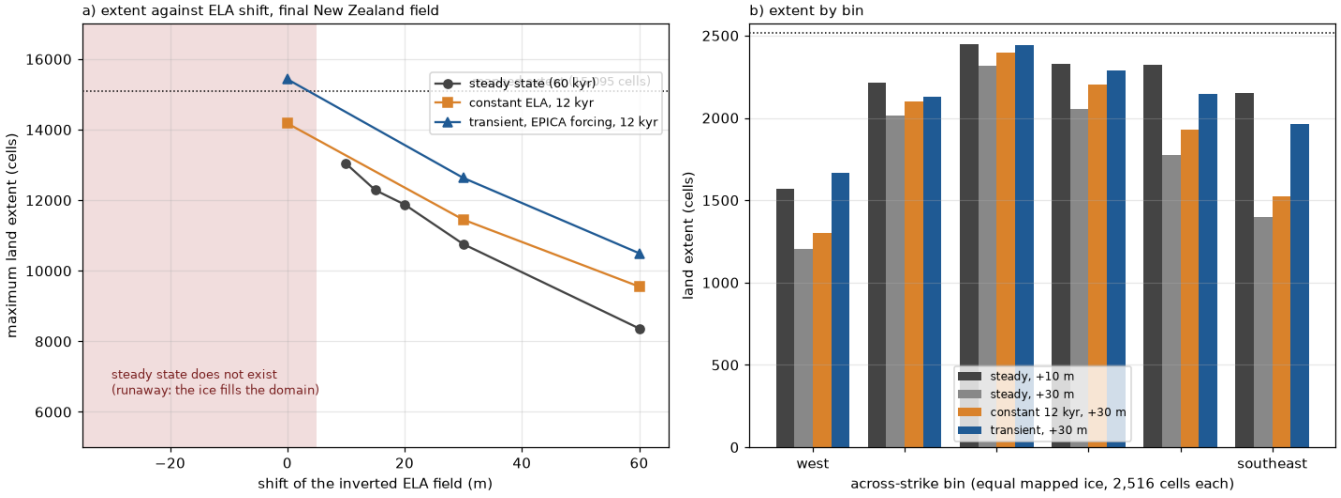


Figure 10. Steady state against transience for the final New Zealand field in IGM. (a) Maximum land extent as a function of a uniform shift δ of the inverted ELA, for a steady state (60 kyr), a constant ELA (12 kyr) and the transient EPICA forcing (12 kyr); the dotted line is the mapped extent (15,095 cells). Steady runs at $\delta \leq 0$ run away and fill the domain (shaded). (b) Land extent in six across-strike bins of equal mapped ice (2,516 cells each), from west to southeast.

3.6 Climate

We start by report the inferred temperature and precipitation anomalies at the three sites (Mt Irene in the wet southwest, Tasman in the centre, South Cameron in the dry east). The intersection of the ΔP–ΔT curve with the Clausius–Clapeyron relation (Fig. 11) gives, with the total uncertainty, a cooling of 5.6 K (4.4 to 6.9) and a precipitation change of -33 % at Mt

Irene, 4.8 K (3.8 to 6.1) and −29 % at Tasman, and 6.5 K (5.1 to 8.1) and −37 % at South Cameron. At unchanged precipitation the cooling is 4.9, 4.1 and 5.5 K. The extent alone fixes only the curve. Along it, the cooling changes from 7.9, 7.3 and 8.7 K for 80 % less precipitation to 4.1, 3.3 and 4.6 K for 50 % more.

The total uncertainty of the cooling at the intersection point, as a standard deviation, is 0.7 to 0.9 K at the three sites. The largest term is the lapse rate (0.5 to 0.65 K for 5 to 7 K km^{-1}), followed by the error of the modern ELA (0.3 K). The spread between the accepted members (0.14 to 0.22 K) is similar to that of the Clausius–Clapeyron constant (0.14 to 0.19 K), the surrogate bias (0.1 to 0.2 K), the seasonality scale (0.1 to 0.2 K) and the elevation profile of the precipitation (0.03 to 0.18 K), and it is smaller than the change in the half-range at South Cameron (0.46 K). The precipitation scale matters little (0.1 K or less).

The restriction to the gradients allowed by the degree-day model matters for the result. Using all 165 accepted members, without the restriction, gives a cooling of 5.9 (4.6 to 7.3), 5.6 (4.3 to 7.1) and 7.1 K (5.5 to 8.9) at the three sites, which is 0.3 to 0.8 K more, and the spread between members is about twice as large (0.35 to 0.42 K against 0.14 to 0.22 K).

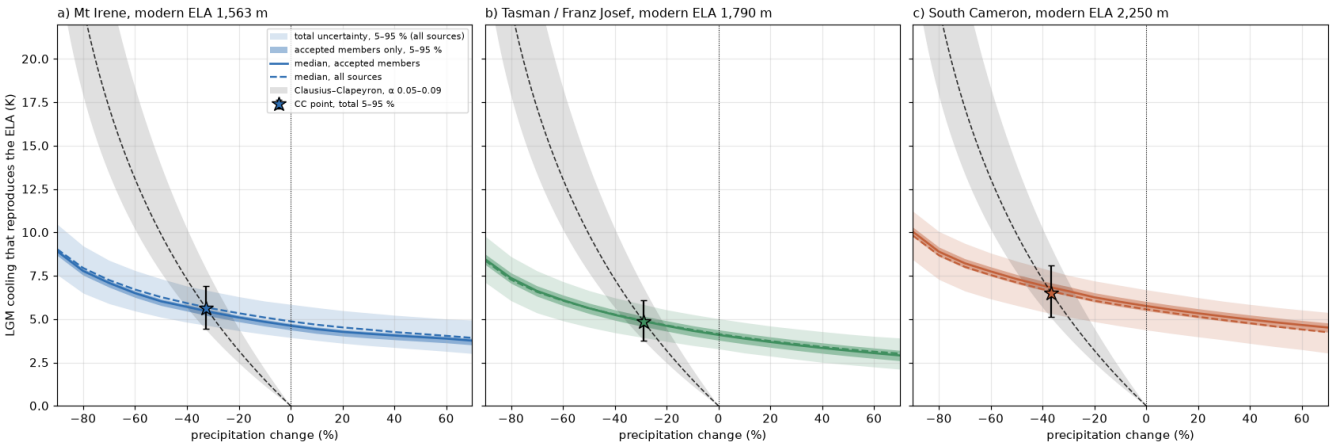


Figure 11. Inferred relationship between cooling and precipitation change at (a) Mt Irene, (b) Tasman and (c) South Cameron. Each curve is the cooling that reproduces the LGM ELA of the inversion at the site, for a given precipitation change, with the stationary temperature calibrated to the modern ELA. The solid lines are the median of the accepted members (the 20 members whose mass-balance gradients lie in the range allowed by the degree-day model), with their 5 to 95 % range as the dark band. The light band is the 5 to 95 % range of all uncertainty sources, which are the choice of member, the lapse rate, the change and the scale of the seasonal half-range, the precipitation, the elevation profile of the precipitation, the Clausius–Clapeyron constant, the bias of the surrogate ELA and the error of the modern ELA (500 draws), and the dashed line is its median. The grey band is the precipitation change implied by the Clausius–Clapeyron relation for $\alpha = 0.05$ to 0.09 K^{-1} and the dashed black line is $\alpha = 0.07$ K^{-1}. The stars are the points where the curve meets the Clausius–Clapeyron line, with their total 5 to 95 % range.

Inside the reach of the modern-ELA data, the inversions lower the ELA by a median of 699 m (525 to 1,023 m, 5 to 95 % over the cells), from a modern median of 1,895 m to 1,169 m. The maps (Fig. 12) give a cooling of 5.1 K (3.7 to 7.3 K) and a precipitation reduction of 30 % in the same area, with the largest cooling in the southwest. The median spread between the accepted members is 73 m for the ELA and 0.5 K for the cooling.

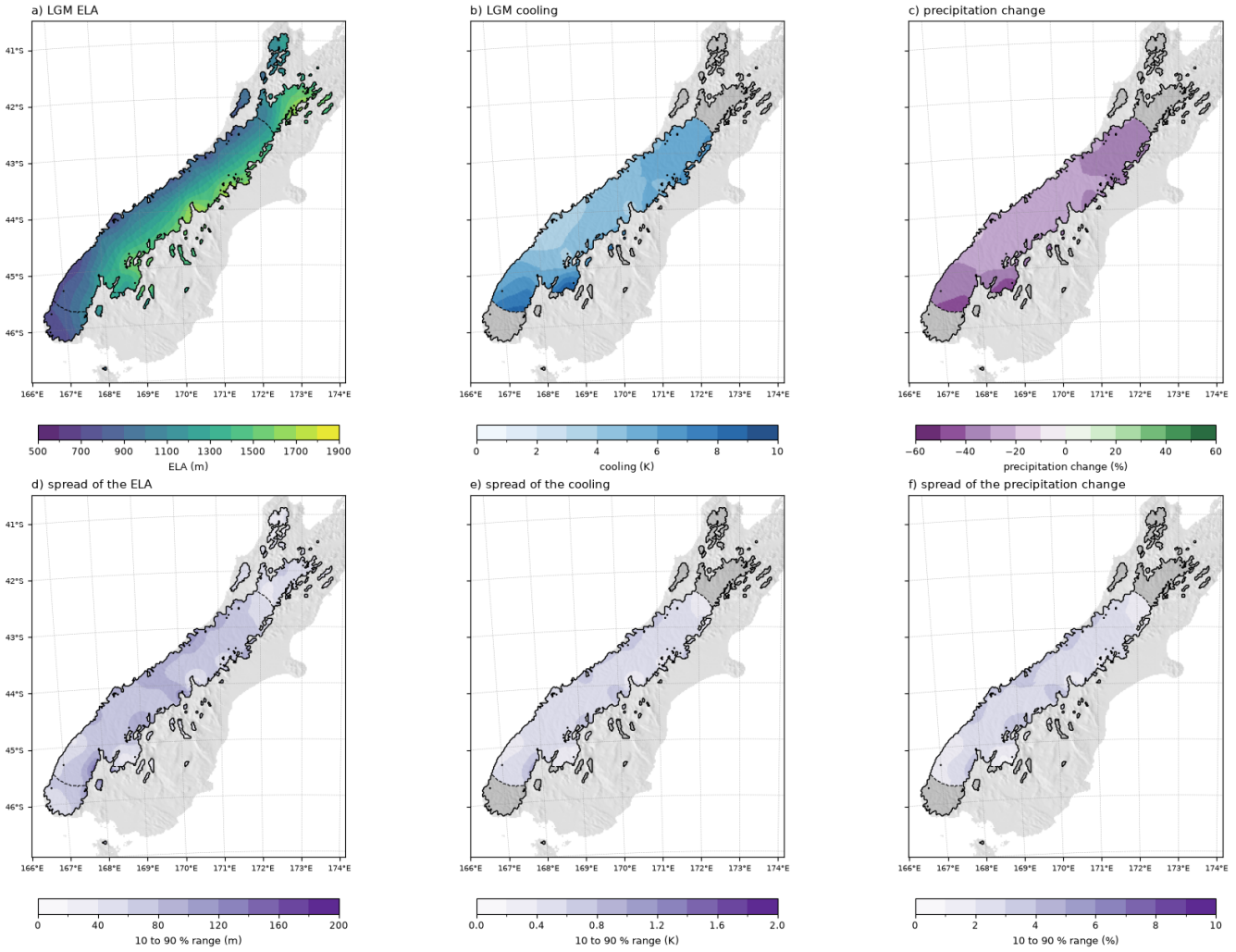


Figure 12. Last Glacial Maximum equilibrium line altitude (ELA), cooling and precipitation change over the mapped ice of the South Island of New Zealand, with the spread between the accepted inversions. (a) ELA, (b) cooling at the ELA and (c) precipitation change, all as the median of the 20 accepted members of the surrogate ensemble whose mass-balance gradients lie in the range allowed by the degree-day model. (d–f) The spread of the same three quantities between the members, given as the difference between the 90th and the 10th percentile. The cooling and the precipitation change are obtained cell by cell from the ELA with the positive-degree-day model, calibrated to the modern ELA of each cell, a lapse rate of $6\,^{\circ}\mathrm{C\,km^{-1}}$ and the Clausius–Clapeyron relationship ($\alpha = 0.07\,\mathrm{K^{-1}}$). The solid black line is the mapped limit of the ice. The dashed line marks the 60 km reach of the glaciers used for the plane. Beyond it the modern ELA is an extrapolation, and the cooling and the precipitation change are not interpreted (grey). The spread in (d–f) includes only the difference between the accepted members, and not the other sources of uncertainty, which are shown in Figs. 11.

4 Discussion

Building on the recent work of Višnjević et al. (2018) and Lleshi et al. (2025), we have presented an inversion scheme that infers past temperature and precipitation from observed ice extents. We first discuss the computational cost of the method, then the choices and limitations that matter when interpreting the results, and finally how the inferences can be compared with independent proxies and climate models.

The cost of the method is dominated by the training data of the surrogate. For New Zealand, one transient IGM run takes about 8 min on an RTX 4090 and about 65 min on a TITAN X. An inversion directly in IGM needs 30 to 40 such runs, about 5 h, so that 400 inversions would take about 2,000 h. The surrogate was trained on 2,052 usable runs out of about 3,600 attempted, which cost about 370 GPU-hours (in RTX 4090 time), plus 75 min of training. After this, one inversion takes about 20 s and 400 inversions take about one hour. The surrogate therefore pays off after roughly 75 inversions, and the sweeps over gradients, amplitudes, forcings and random starts used here require many more. For a single inversion, IGM alone is cheaper. The SIA is fast (4 min per field) and serves only to start the iteration and to guide the first round of runs. The cost grows with the number of cells and with the time step that the ice requires, so it will increase for larger domains and thicker ice caps, which may also need finer resolution. This additional cost depends on the domain and must be assessed case by case. The surrogate is also specific to a given forcing, time window and initial ice configuration, and testing other choices requires more IGM runs or a new surrogate.

The extent of a glacier at a given ELA is not set by the climate alone. It also depends on the time available for the ice to grow and on the ice it starts from, and this distinction is central to the choice between steady-state and transient inversions (Višnjević et al., 2018; Lleshi et al., 2025). A steady-state inversion finds the ELA at which the modelled glacier reaches the observed extent at equilibrium, whereas a transient inversion finds the ELA that reaches it within a given period under a given forcing. In IGM the two differ for our final field (Fig. 10). Under the transient forcing the field reproduces the mapped extent (15,429 cells against 15,095); held at a constant ELA it gives 14,178 cells after 12 kyr and less at equilibrium. The steady response to the ELA is steep exactly where the LGM extent sits, and it ends in a fold: for a shift of zero or less, there is no steady state and the ice fills the domain. A steady-state inversion is therefore ill-conditioned there. In a transient inversion, the ice has only a finite time and does not run away within the window (15,429 cells at $\delta = 0$).

The sign of the bias of a steady-state inversion is not general. For this field, steady states have less ice than the transient run, and the ELA of a steady-state inversion would be about 25 to 28 m too low, because the transient forcing is colder than its mean at the peak and the steady runs are still losing the starting ice. For the SIA field, which is not a transient solution, the bias goes the other way. Held at a constant ELA its extent grows from 14,710 cells to 18,691 after 3 kyr and 19,531 at equilibrium (IGM, 20 kyr), so a steady-state inversion would return an ELA that is too high and a cooling that is too small. Whether it is too high or too low therefore depends on how the ice and the climate it has seen relate to equilibrium. A transient inversion makes this explicit through the initial ice, the duration and the forcing, and constrains the time available for growth, rather than assuming that the mapped extent is an equilibrium response to the climate. The SIA and IGM give similar ice over land under the same forcing, so the limitation lies in the scheme in which the ELA and the ice extent evolve together, not in the forward model.

The inversion of the extents for the ELA does not give a temperature and a precipitation change separately. It gives, at each location, a curve of the combinations that reproduce the mapped limit. Across the three South Island sites, the cooling ranges from 3.3 K for 50 % more precipitation to 8.7 K for 80 % less precipitation (Fig. **??**). The extent alone does not select a point on this curve, and the Clausius–Clapeyron relation offers a natural reference, since it links the two variables through the response of atmospheric moisture to temperature alone. It is nevertheless an assumption, and we prefer to use it as a benchmark rather than as a closure. An independent estimate of either the cooling or the precipitation change, from a proxy or from a model, then fixes a point on the curve. A departure of this point from the Clausius–Clapeyron value would not be an error of the method but potentially a measure of what the thermodynamic argument leaves out, such as a change in the position of the storm tracks, in the flow over the orography or in the share of snow in the precipitation. In this sense the ELA is a quantity that no assumption about the climate has to precede, and a single independent measurement turns it into a test of atmospheric dynamics.

The largest uncertainties in the cooling come from the lapse rate (0.5 to 0.65 K) and from the error of the modern ELA (0.3 K), while the other sources contribute 0.1 to 0.2 K each. The change in seasonality matters most in the dry east (0.46 K at South Cameron), and the elevation profile of the precipitation shifts the cooling by up to ± 0.3 to 0.7 K, depending on where the modern ELA lies relative to the level of maximum precipitation. A larger, structural uncertainty comes from the choice of the mass-balance gradients. Restricting them to the values permitted by the degree-day model reduces the cooling by 0.3 to 0.8 K and halves the spread among the accepted inversions. The elevation at which the cooling is reported must also be specified whenever the lapse rate is allowed to vary. Because the glacier inversion is separated from the climate calculation, these assumptions can be changed and the cooling recomputed without repeating the inversions.

Our results are conditional on the forcing, the initial ice and the mass-balance assumptions used in the inversion. For New Zealand, the inferred extent and cooling are relatively insensitive to the forcing history. Reducing the amplitude of the EPICA index from 900 to 500 m, shifting its peak by ± 2 kyr or removing its variability changes the across-strike gradient by at most 4 m per 100 km, the mean ELA by about 50 m and the cooling by 0.3 K. Starting the window 0.5 to 14.5 kyr earlier changes the ELA field by 11 to 29 m (root mean square), and starting the run from no ice or from half of the SIA ice changes it by less than 1 m. This insensitivity reflects the small variations of the index over the LGM window (it stays between 0.91 and 1.06 of its glacial mean) and should not be assumed elsewhere. Northern regions may need a different forcing history, and an Antarctic record may not capture the timing or amplitude of the temperature changes in the Alps as it does in New Zealand. The forcing amplitude is also not constrained independently by the index, and moraines from different lobes may be asynchronous. Comparing maximum extents over a time window accommodates this asynchrony, but it cannot tell when each lobe reached its maximum.

Other assumptions remain to be tested across regions. For New Zealand, moving to a 1 km grid changes the extent by 5 %, equivalent to about 6 m of ELA. The surrogate agrees closely with IGM, although its training does not cover the full range of ablation gradients allowed by the degree-day model. Precipitation and the half-range of the seasonal cycle at the key locations are prescribed, and the modern ELA is derived from a plane fitted to the regional glaciers. The residual of this fit is 146 m, equivalent to about 0.9 K at unchanged precipitation, so we do not interpret results far from these glaciers. A single pair of mass-balance gradients across the domain is also a simplification, since the degree-day model implies variations of 40 to 70 %

across New Zealand, and the effect of letting them vary in space is untested. The ice model neglects calving and treats the lithosphere as an elastic plate.

The results can finally be set against independent records. Pollen-based reconstructions of the Last Glacial Maximum (Cleator et al., 2020) give a cooling of 3.0 to 3.3 K in annual mean temperature in the two grid cells covering the South Island, with the coldest month cooling by 3.5 to 4.2 K and the warmest month by 2.5 to 2.6 K, each with an uncertainty of about 1 K. These values describe lowland vegetation, whereas our estimate refers to the altitude of the equilibrium line, so the glaciers are expected to record a larger cooling if the lapse rate was steeper than today or if the lowlands were buffered by the ocean. The assimilation of sea-surface proxies in a climate model (Tierney et al., 2020) gives a cooling of 5.2 K for the waters around New Zealand and 6.2 K for the air over the region, which falls within our values of 4.1 to 5.5 K without a change in precipitation and 4.8 to 6.5 K with Clausius–Clapeyron scaling. The air cools by about 1 K more than the sea surface, a first indication of the land-sea contrast that the glaciers should record. Our precipitation change of -27 to -37 % exceeds the reduction of up to 25 % in the forward model of Golledge et al. (2012), in line with our larger cooling, since the two trade against each other along the curve in Fig. **??**. The models of the fourth phase of the Paleoclimate Modelling Intercomparison Project (Kageyama et al., 2021) for which we could obtain both a Last Glacial Maximum and a control run give a cooling of 0.7 to 2.8 K over the South Island and the surrounding sea, and a change in precipitation between -10 and $+2$ %. Both are smaller than our values, which agrees with the tendency of models to underestimate regional cooling and precipitation change noted in the introduction, although two of the four runs are short and may not be in equilibrium.

The seasons provide a further comparison. Because melt during the warm season sets the ELA, the glacier extent constrains the warm-month cooling directly, and the annual cooling follows once the change in the seasonal range is known. The pollen records indicate that the coldest month cooled by 0.9 to 1.7 K more than the warmest month, which implies a half-range increase of about 0.5 to 0.9 K and a warm-month cooling of about 4 to 6 K at the Clausius–Clapeyron point. The models show instead a slightly stronger cooling in summer than in winter. This comparison remains indicative, since the half-range is not known independently and the proxies and models differ in season, elevation and resolution. The same analysis can nevertheless be repeated for each of the mapped regions, and a joint analysis of ice extents, pollen, sea-surface records and climate models would show whether a single set of changes in temperature, precipitation and seasonality can explain them at the global scale.

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